



Cloud Computing Essentials

Pradnya Garde

Associate Professor

Department of Computer Applications

Cloud Computing Essentials

Introduction To Cloud Computing

Pradnya Garde

Associate Professor

Department of Computer Applications

Cloud Computing Essentials

Introduction to Cloud Computing – Concepts and Technology



What is Cloud Computing?

- Cloud computing is a scalable services consumption and delivery platform that provides on-demand computing service for shared pool of resources, namely servers, storage, networking, software, database, applications etc., over the Internet.(Generic)
- Cloud computing delivers on-demand IT resources—such as servers, storage, databases, and software—over the internet with a pay-as-you-go pricing model.(AWS)

Cloud Computing Essentials

Introduction to Cloud Computing – Concepts and Technology



Why Cloud Computing?

- Trade fixed expense for variable expense
- Benefit from massive economies of scale
- Stop guessing capacity
- Increase speed and agility
- Stop spending money running and maintaining data centers
- Go global in minutes

Cloud Computing Essentials

Introduction to Cloud Computing – Concepts and Technology



Characteristics of Cloud Computing

- On-demand Self-Service - Users can provision resources (servers, network) as needed automatically
- Broad Network Access - Services are accessible over the internet from various devices (mobile, laptops),
- Resource Pooling - Providers serve multiple consumers using a multi-tenant model, Rapid Elasticity),
- Highly Available and Scalable - Resources-redundancy, fail-overs, geographically separated, scale on-demand
- Professionally Secured Infrastructure
- Metered Payment Model - Usage is monitored, and customers pay only for what they use

Cloud Service Models

Cloud services are structured into three main models, defining the level of management responsibility.

- **IaaS (Infrastructure as a Service)**
- **PaaS (Platform as a Service)**
- **SaaS (Software as a Service)**

Cloud Computing Essentials

Introduction to Cloud Computing – Concepts and Technology



IaaS (Infrastructure as a Service)

- Provides virtualized computing resources (virtual machines, storage, networking) over the internet.
- It offers maximum control over operating systems and infrastructure.
Examples: Microsoft Azure, Google Cloud Compute Engine.
- **Real Life Analogy:** You rent an apartment, an empty building shell. You are responsible for everything inside the house (furniture = applications, decorations = data, lighting = operating system), while the landlord only cares about the structure.
- **AWS IaaS products** include Elastic Cloud Compute (EC2) for virtual machine instances, Elastic Block Store (EBS) for storage volumes, and Elastic Load Balancing

Cloud Computing Essentials

Introduction to Cloud Computing – Concepts and Technology



PaaS (Platform as a Service)

- Offers hardware and software tools—usually for application development—over the internet.
- Developers can build, test, and deploy applications without managing underlying infrastructure. *Examples: Heroku, Google App Engine.*
- **Real Life Analogy:** You rent a furnished apartment. The structure, plumbing, electricity, and basic furniture are already in place. You don't care about repairing the plumbing or building the walls (OS/servers), but you choose how to arrange the furniture and manage your personal belongings (your code/data).
- **AWS PaaS products** include Elastic Beanstalk and Elastic Container Service (ECS).

Cloud Computing Essentials

Introduction to Cloud Computing – Concepts and Technology



SaaS (Software as a Service)

- Delivers complete, ready-to-use software applications via a web browser or app.
- The provider manages everything, from infrastructure to updates.
Examples: Gmail, Salesforce, Microsoft Office 365.
- **Real Life Analogy:** You stay in a hotel. Everything is fully furnished and taken care of (cleaning, utilities, security). The landlord manages everything. You only bring your suitcase (data/user settings) and use the room as it is.

Cloud Computing Essentials

Introduction to Cloud Computing – Concepts and Technology



Cloud Service Models

The breakdown of responsibility across multiple infrastructure types:

	On-premises Deployments	IaaS	PaaS	SaaS
Your Responsibility	Application code	Application code	Application code	Application code
	Security	Security	Security	Security
	Database	Database	Database	Database
	OS	OS	OS	OS
	Virtualization	Virtualization	Virtualization	Virtualization
	Networking	Networking	Networking	Networking
	Storage hardware	Storage hardware	Storage hardware	Storage hardware
	Server hardware	Server hardware	Server hardware	Server hardware
Cloud Platform Responsibility				

Cloud Computing Essentials

Introduction to Cloud Computing – Concepts and Technology



Cloud Applications

- Cloud applications in cloud computing are software programs hosted on remote servers and accessed via the internet, rather than being stored on a local device.
- They offer high scalability, on-demand, and multi-tenant access, commonly delivered as Software as a Service (SaaS).
- Cloud computing supports diverse, modern applications across various sectors:
 - Data Storage & Backup, Big Data Analytics, Business Applications, Communication and Collaboration etc.

Cloud Computing Essentials

Introduction to Cloud Computing – Concepts and Technology



Cloud Applications – Examples

- **Data Storage & Backup:** Storing data securely off-site (e.g., Dropbox, Google Drive, One Drive).
- **Application Development, AI & Testing:** Using PaaS to quickly set up environments. Platforms offering AI/ML capabilities, such as AWS, Google Cloud, or Microsoft Azure
- **Big Data Analytics:** Processing large datasets using scalable cloud resources.
- **Business Applications:** Using CRM (Salesforce) , accounting (QuickBooks), HR tools and ERP systems.
- **Entertainment, Content Delivery & Streaming:** Delivering media content efficiently. Platforms like Netflix, Spotify, and gaming services.
- **Communication and Collaboration:** Tools like Slack, WhatsApp, and Microsoft Teams.

Cloud Computing Essentials

Introduction to Cloud Computing – Concepts and Technology



Core Concepts and Technology

- Virtualization
- Load Balancing
- Scalability
- Elasticity

Cloud Computing Essentials

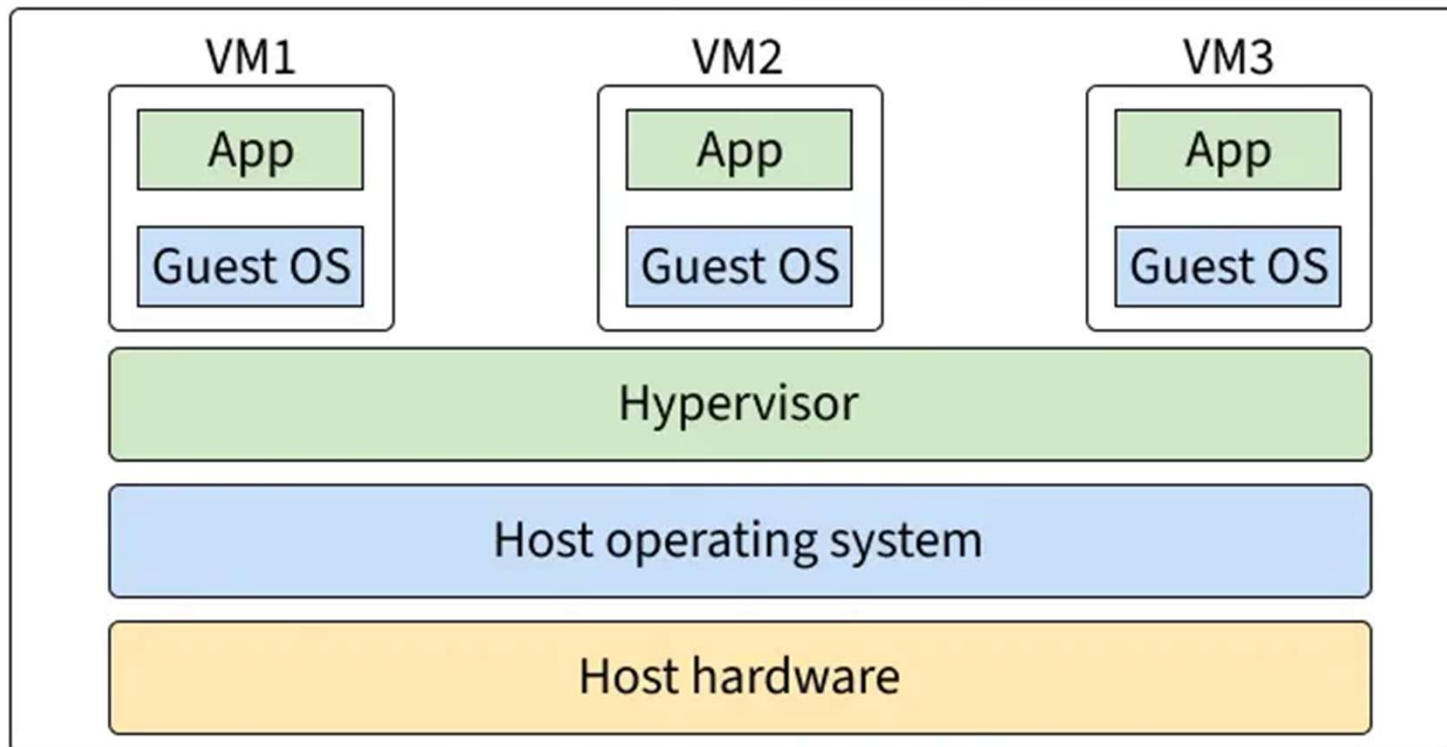
Introduction to Cloud Computing – Concepts and Technology



Virtualization

- The foundational technology of cloud computing
- Enables the creation of multiple simulated environments or Virtual Machines (VMs) from a single physical hardware system
- AWS carves the necessary resources from larger existing devices.

Virtual machines

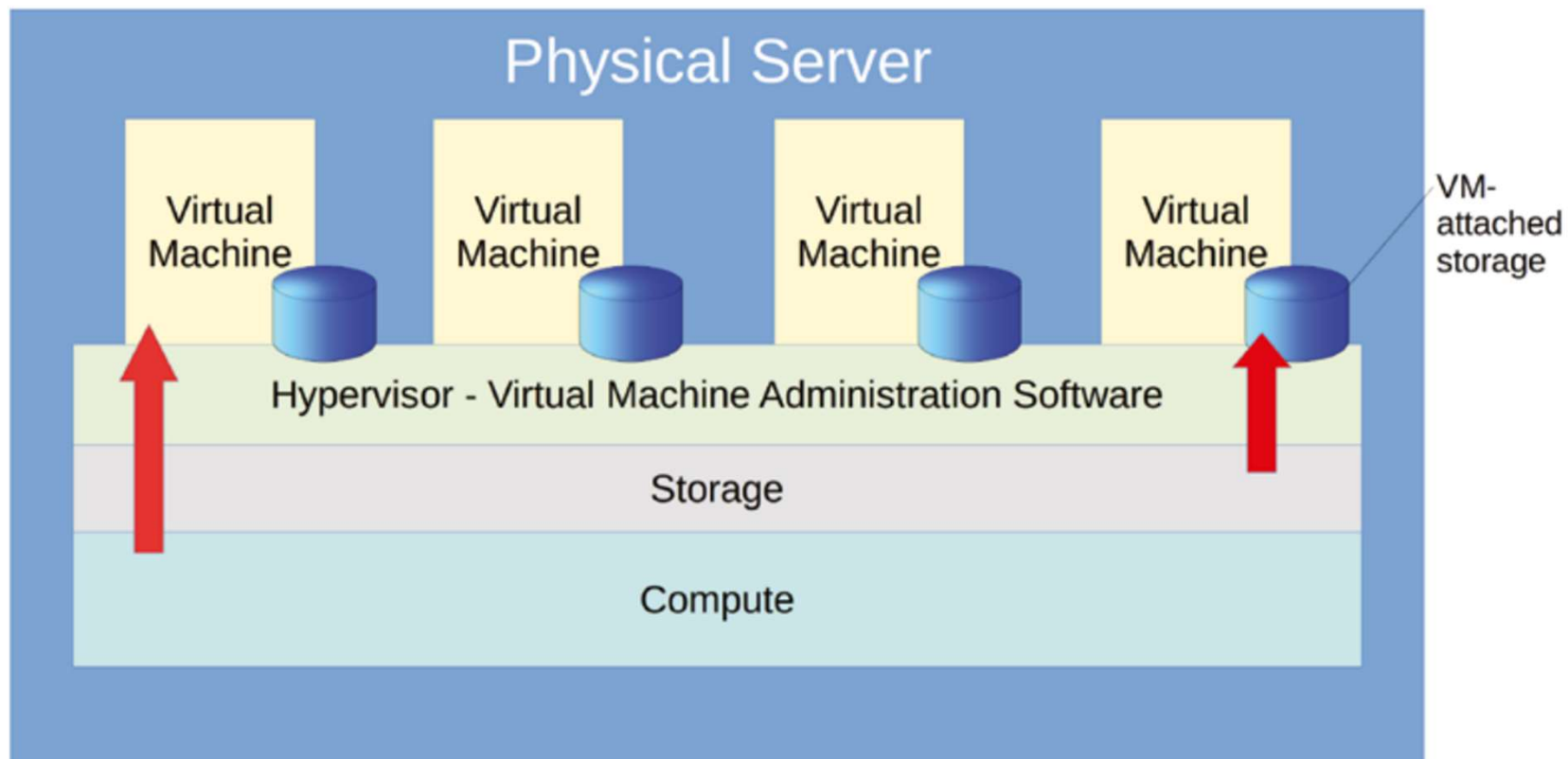


Virtualization - Key Aspects

- **Core Functionality:** Virtualization decouples software from hardware, allowing multiple operating systems (OS) and applications to run simultaneously on one physical machine, with each VM operating independently.
- **Hypervisor Technology:** A hypervisor (or Virtual Machine Monitor) manages the virtualized environment, allocating physical resources—memory, CPU, and storage—to each VM.

Cloud Computing Essentials

Introduction to Cloud Computing – Concepts and Technology



Virtualization - Types of Virtualization:

- **Server Virtualization:** Partitioning a physical server into multiple virtual servers.
- **Storage Virtualization:** Pooling physical storage from multiple network devices into what appears to be a single storage unit.
- **Network Virtualization:** Combining available resources in a network by splitting up the available bandwidth into independent channels.
- **Application Virtualization:** Allowing users to run applications without installing them directly on the operating system.

Cloud Computing Essentials

Introduction to Cloud Computing – Concepts and Technology



Virtualization - Benefits

- **Efficiency:** Improved utilization of hardware resources.
- **Cost Savings:** Lower IT, power, and maintenance costs.
- **Flexibility & Agility:** Rapid deployment of new services and workloads.
- **Resiliency:** Enhanced disaster recovery and reduced downtime.

Virtualization allows providers like [AWS](#), [Microsoft Azure](#), and [IBM](#) to offer flexible, scalable cloud services.

Cloud Computing Essentials

Introduction to Cloud Computing – Concepts and Technology



Load balancing

- In cloud computing, it is a software-defined technique that distributes incoming application traffic and workloads across multiple servers, virtual machines, or data centers (pool of resources).
- It optimizes resource utilization and enables high-availability by rerouting traffic during server failures.
- Load balancing is essential for modern, high-traffic applications, acting as a traffic cop to ensure efficient data flow and optimal user experience.

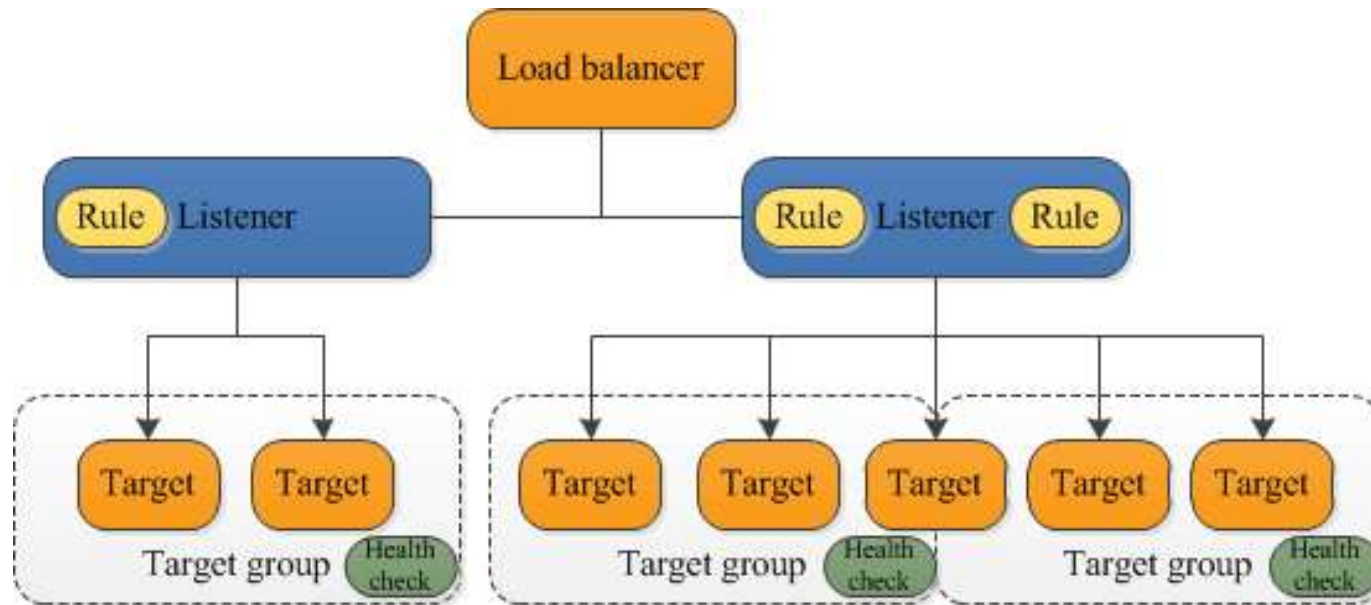
Load balancing - Key Aspects

A load balancing algorithm is the set of rules that a load balancer follows to determine the best server for each of the different client requests. Load balancing algorithms fall into two main categories.

- **Static load balancing** - follow fixed rules and are independent of the current server state. The following are examples of static load balancing. *Example: Round Robin (sequential), and IP Hash (based on client IP).*
- **Dynamic load balancing** - examine the current state of the servers before distributing traffic. The following are some examples of dynamic load balancing algorithms. *Example: Least Connections (directs to the server with fewest active connections).*

Cloud Computing Essentials

Introduction to Cloud Computing – Concepts and Technology



Load balancing – Types

Load balancing can be classified into main categories based on the layer of the OSI model at which they operate and what information they inspect in the client request to redirect traffic. The three primary types are:

- **Layer 3 (Network Layer):** Balances traffic based on IP packets. *Ex: Gateway Load Balancer (GLB)*
- **Layer 4 (Transport Layer):** Balances traffic based on data from network protocols like TCP or UDP. *Ex: Network load balancer (NLB), Classic Load Balancer (CLB)*
- **Layer 7 (Application Layer):** Routes traffic based on application-level information, such as HTTP/HTTPS headers, URLs, or cookies. *Ex: Application load balancer (ALB)*

Cloud Computing Essentials

Introduction to Cloud Computing – Concepts and Technology



Load balancing - Benefits

- **Scalability:** Allows adding or removing servers based on demand.
- **Reliability & Availability:** Detects server health and directs traffic only to healthy servers.
- **Reduced Costs:** Often provided as-a-service (LBaaS), eliminating the need for expensive hardware maintenance.

Cloud Computing Essentials

Introduction to Cloud Computing – Concepts and Technology



Scalability

- A scalable service will automatically grow in capacity to seamlessly meet any changes in demand.
- A well-designed cloud-based operation will constantly monitor the health of its application stack and respond whenever preset performance metrics might soon go unmet.
- The response might include automatically launching new server instances to add extra compute power to your existing cluster.
- But it will probably also involve prepopulating those instances with the application data and configuration settings

Cloud Computing Essentials

Introduction to Cloud Computing – Concepts and Technology



Elasticity

- You can stretch an elastic band far beyond its resting state. But part of what makes it truly elastic is the fact that, when you let go of it, it immediately returns to its original size.
- The reason the word elastic is used in the names of so many AWS services (Elastic Compute Cloud, Elastic Load Balancing, Elastic Beanstalk, and so on) is because those services are built to be easily and automatically resized.
- Generally, you set the maximum and minimum performance levels you want for your application, and the AWS service(s) you're using will automatically add or remove resources to meet changing usage demands.



THANK YOU

Pradnya Garde

Associate Professor

Department of Computer Applications

pradnyagarde@pes.edu



Cloud Computing Essentials

Pradnya Garde

Associate Professor

Department of Computer Applications

Cloud Computing Essentials

Cloud Architecture

Pradnya Garde

Associate Professor

Department of Computer Applications

What is Cloud Computing Architecture?

- Architecture of cloud computing is the combination of both SOA (Service Oriented Architecture) and EDA (Event Driven Architecture)
- The cloud architecture is divided into 2 parts:
 - Frontend
 - Backend

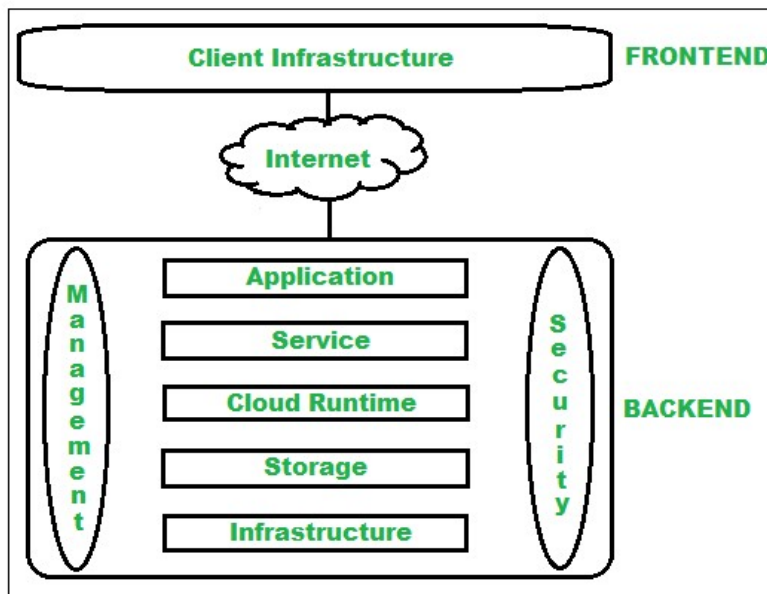
Key Components of Cloud Architecture

- **Frontend:** The client-side interface (web browsers, mobile apps, applications) that users interact with to access the cloud.
- **Backend:** The cloud provider's infrastructure (servers, data storage, virtual machines) that manages and delivers services.
- **Cloud-Based Delivery:** The mechanism, including networking and the internet, that connects the front end and back end.
- **Middleware:** Software that connects different components, enabling communication and data management between them.

Cloud Computing Essentials

Cloud Architecture

Key Components of Cloud Architecture



Cloud Deployment Models

- A cloud deployment model fundamentally determines where the infrastructure for your deployment exists and who owns and controls that infrastructure.
- It identifies the specific type of cloud environment based on ownership, scale, and access, as well as the cloud's nature and purpose.
- It specifies the location of the servers utilized. It specifies how your cloud infrastructure will look, what you can change, and whether you will be given services or will have to create everything yourself.

Cloud Deployment Models

- **Public Cloud:** Resources owned and operated by a third-party provider, shared across multiple tenants. Minimal Investment, No maintenance, Less secure, Low customization
- **Private Cloud:** Resources used exclusively by one organization, offering higher customization and security. Better control, Higher security, Less scalable, Costly
- **Hybrid Cloud:** A combination of public and private clouds, allowing data and applications to be shared between them. Flexible, Secure, Difficult to manage, slow transmission

Cloud Deployment Models

- **Community Cloud:** It is a distributed system that is created by integrating the services of different clouds to address the specific needs of a community, industry, or business. Cost-effective, Better security, Limited scalability, Rigid customization
- **Multi-Cloud:** Uses many public clouds. Reduced latency, High availability, Complex, Security issues.

Cloud Deployment Models - Comparison

Factors	Public Cloud	Private Cloud	Community Cloud	Hybrid Cloud
Initial Setup	Easy	Complex, requires a professional team to setup	Complex, requires a professional team to setup	Complex, requires a professional team to setup
Scalability and Flexibility	High	High	Fixed	High
Cost-Comparison	Cost-Effective	Costly	Distributed cost among members	Between public and private cloud
Reliability	Low	Low	High	High
Data Security	Low	High	High	High
Data Privacy	Low	High	High	High



THANK YOU

Pradnya Garde

Associate Professor

Department of Computer Applications

pradnyagarde@pes.edu



Cloud Computing Essentials

Pradnya Garde

Associate Professor

Department of Computer Applications

Cloud Computing Essentials

Cloud Architecture

Pradnya Garde

Associate Professor

Department of Computer Applications

What is Cloud Computing Architecture?

- Architecture of cloud computing is the combination of both SOA (Service Oriented Architecture) and EDA (Event Driven Architecture)
- The cloud architecture is divided into 2 parts:
 - Frontend
 - Backend

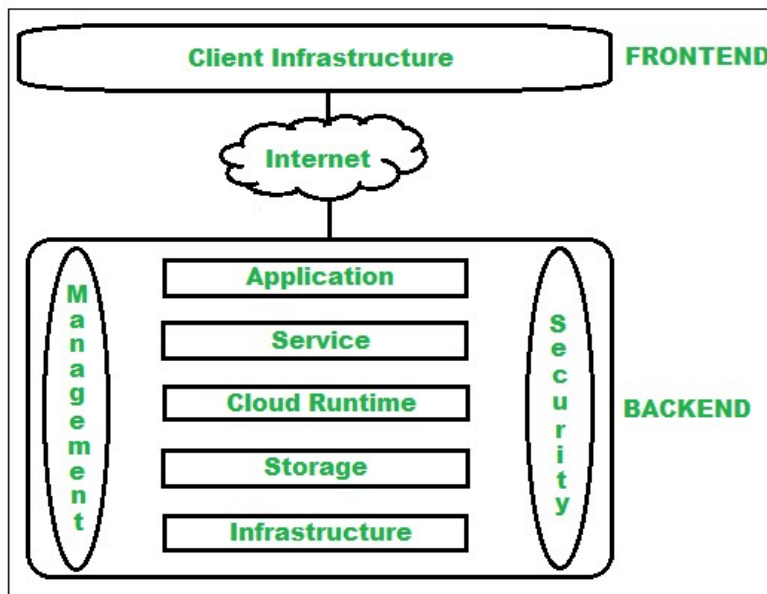
Key Components of Cloud Architecture

- **Frontend:** The client-side interface (web browsers, mobile apps, applications) that users interact with to access the cloud.
- **Backend:** The cloud provider's infrastructure (servers, data storage, virtual machines) that manages and delivers services.
- **Cloud-Based Delivery:** The mechanism, including networking and the internet, that connects the front end and back end.
- **Middleware:** Software that connects different components, enabling communication and data management between them.

Cloud Computing Essentials

Cloud Architecture

Key Components of Cloud Architecture



Cloud Deployment Models

- A cloud deployment model fundamentally determines where the infrastructure for your deployment exists and who owns and controls that infrastructure.
- It identifies the specific type of cloud environment based on ownership, scale, and access, as well as the cloud's nature and purpose.
- It specifies the location of the servers utilized. It specifies how your cloud infrastructure will look, what you can change, and whether you will be given services or will have to create everything yourself.

Cloud Deployment Models

- **Public Cloud:** Resources owned and operated by a third-party provider, shared across multiple tenants. Minimal Investment, No maintenance, Less secure, Low customization
- **Private Cloud:** Resources used exclusively by one organization, offering higher customization and security. Better control, Higher security, Less scalable, Costly
- **Hybrid Cloud:** A combination of public and private clouds, allowing data and applications to be shared between them. Flexible, Secure, Difficult to manage, slow transmission

Cloud Deployment Models

- **Community Cloud:** It is a distributed system that is created by integrating the services of different clouds to address the specific needs of a community, industry, or business. Cost-effective, Better security, Limited scalability, Rigid customization
- **Multi-Cloud:** Uses many public clouds. Reduced latency, High availability, Complex, Security issues.

Cloud Deployment Models - Comparison

Factors	Public Cloud	Private Cloud	Community Cloud	Hybrid Cloud
Initial Setup	Easy	Complex, requires a professional team to setup	Complex, requires a professional team to setup	Complex, requires a professional team to setup
Scalability and Flexibility	High	High	Fixed	High
Cost-Comparison	Cost-Effective	Costly	Distributed cost among members	Between public and private cloud
Reliability	Low	Low	High	High
Data Security	Low	High	High	High
Data Privacy	Low	High	High	High



THANK YOU

Pradnya Garde

Associate Professor

Department of Computer Applications

pradnyagarde@pes.edu



Cloud Computing Essentials

Pradnya Garde

Associate Professor

Department of Computer Applications

Cloud Computing Essentials

Introduction to Amazon Web Services (AWS)

Pradnya Garde

Associate Professor

Department of Computer Applications

AWS Shared Responsibility Model

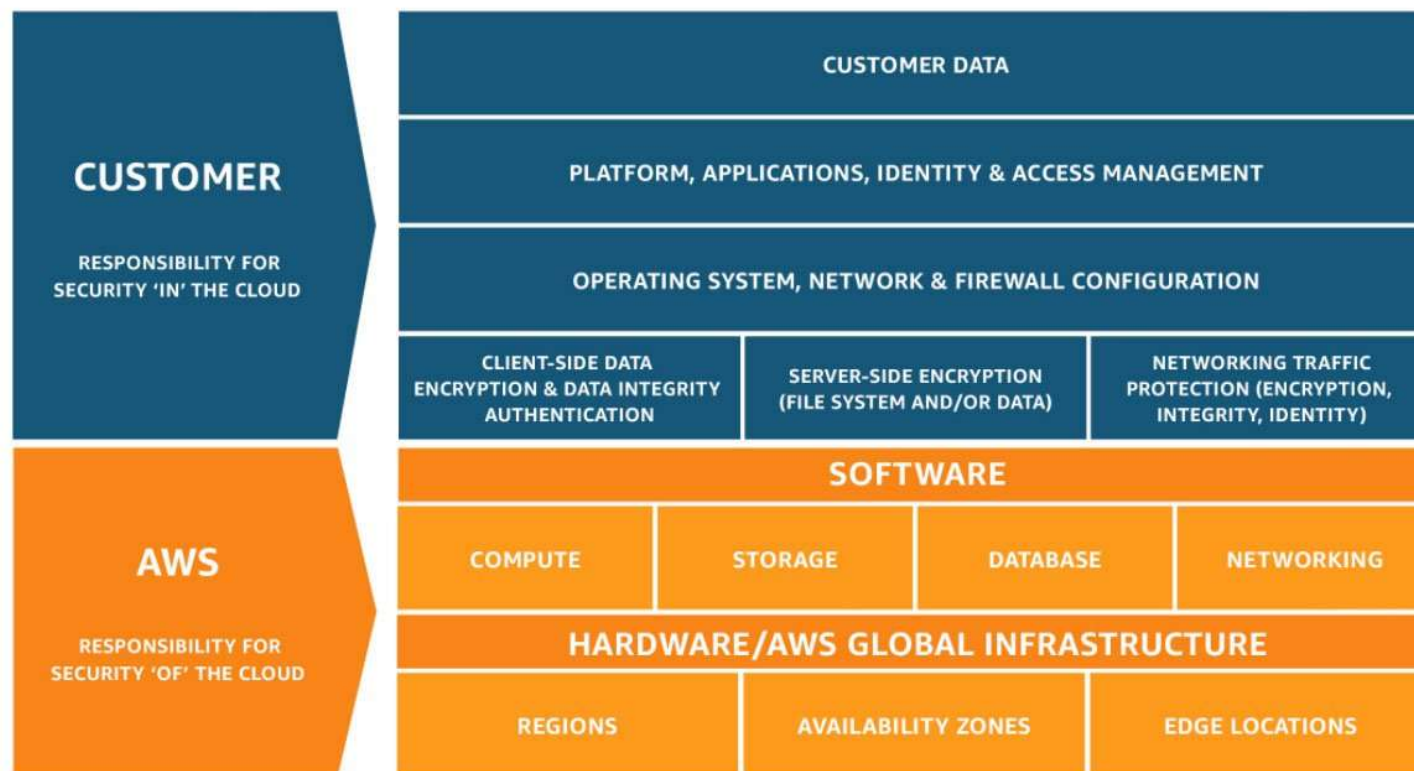
What is a shared responsibility model?

- The shared responsibility model is a **framework that defines the division of cloud security responsibilities between cloud service providers (CSPs) and their customers.**
- The AWS Shared Responsibility Model divides security tasks between AWS and the customer, defined as **"Security of the Cloud" (AWS) and "Security in the Cloud" (Customer).**
- AWS manages physical infrastructure, hardware, and virtualization, while customers are responsible for guest OS patching, firewall configuration, data encryption, and identity management (IAM).

Cloud Computing Essentials

Introduction to Amazon Web Services (AWS)

AWS Shared Responsibility Model



Overview of AWS Management Console

AWS provides three primary methods for interacting with its services, each suited for different use cases:

- the **AWS Management Console** for a visual interface, Graphical User Interface (GUI) via web browser. Programming knowledge not required, point-and-click interface
- the **AWS CLI** for command-line automation, Command-line interface via terminal/shell. Programming knowledge not required, but scripting knowledge (Bash, Python) is needed for automation
- the **AWS SDKs** for integrating AWS into application code, Language-specific libraries for application code. Requires knowledge of a supported programming language (Python, Java, Node.js, etc.)

Overview of AWS Management Console

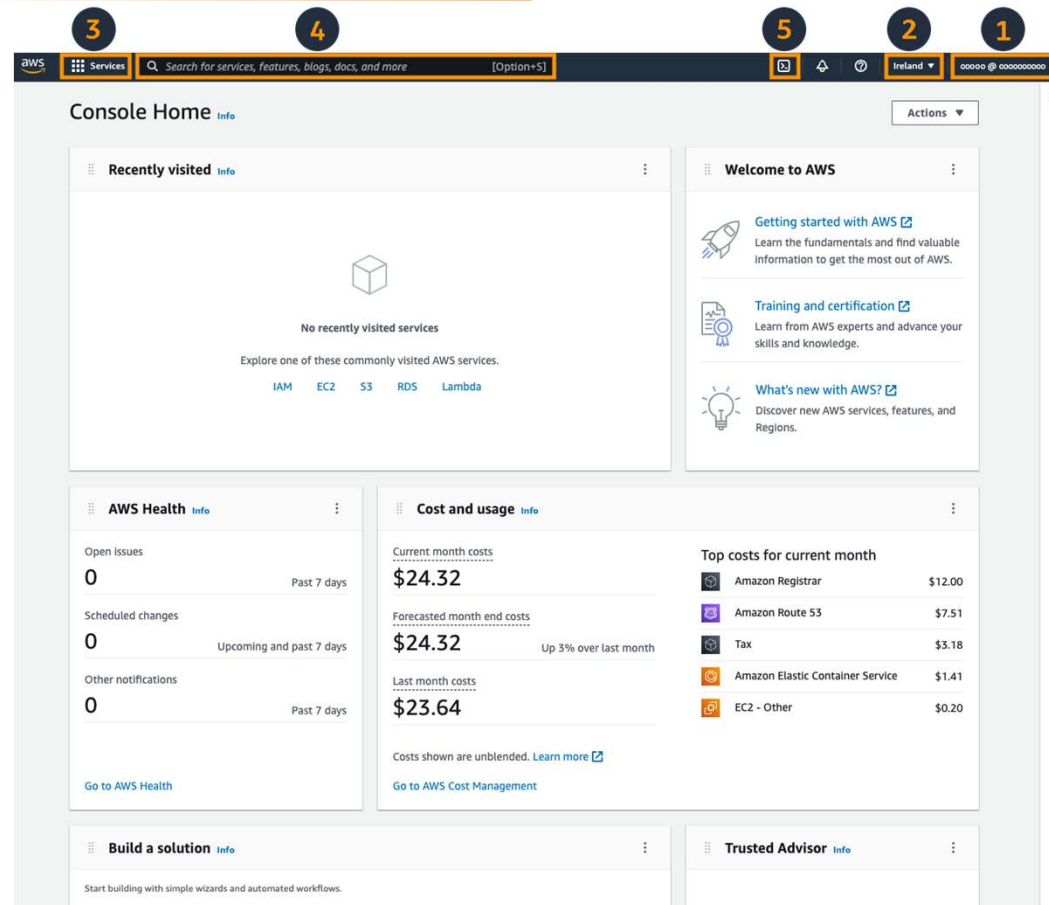
- After signing up for a new AWS account and logging in, you will see the console dashboard.
- This is the starting point for interacting with the various AWS services and other important console components.
- The dashboard consists of a navigation bar at the top and a number of widgets in the main body of the page, which you can configure and rearrange.

Cloud Computing Essentials

Introduction to Amazon Web Services (AWS)

Five highlighted controls in navigation bar:

1. Account information
2. Region selector
3. Service selector
4. Search box
5. AWS CloudShell





THANK YOU

Pradnya Garde

Associate Professor

Department of Computer Applications

pradnyagarde@pes.edu



Cloud Computing Essentials

Pradnya Garde

Associate Professor

Department of Computer Applications

Cloud Computing Essentials

Unit 1 - Experiential Learning

Pradnya Garde

Associate Professor

Department of Computer Applications

Cloud Computing Essentials

Unit 1 - Experiential Learning



- Creating an AWS free-tier account,
- Navigating the AWS Management Console,
- Exploring regions, availability zones,
- and AWS services dashboard

Cloud Computing Essentials

Unit 1 - Experiential Learning



EXERCISE 1.1

Create an AWS Account

1. Go to <https://aws.amazon.com> and click the Create An AWS Account button. If, instead, you see a Sign In button, you might have previously logged in to an existing account using this browser. If you'd still like to create a new account, click Sign In and then create a new, different account.
2. Enter the email address you want to associate with the account as the root user, a name for your AWS account, and a new AWS account name you'd like to use. Select Verify Email Address. You'll need to retrieve the verification code that AWS will send to your email account. You'll then need to create a (strong) password.

..Contd.

Cloud Computing Essentials

Unit 1 - Experiential Learning



EXERCISE 1.1

Create an AWS Account (contd)

3. Select an account type (either Professional or Personal—the only difference is the Company Name field) and the other requested information. Then agree to the AWS terms and select Create Account And Continue.
4. Enter a payment method and, if the payment address is different from the address you used in the previous step, enter a new address and select Secure Submit.
5. On the Verify Your Identity page, enter the contact method AWS can use to send you a verification code. Enter the PIN that you'll receive and continue.
6. One last step: On the Support Plan page, select an AWS support plan. If you're not sure which plan you want, just go with the Basic plan for now. It's free, and you can always upgrade later.



THANK YOU

Pradnya Garde

Associate Professor

Department of Computer Applications

pradnyagarde@pes.edu